Will it fit?

Will this data fit in memory?

Names: _____

Work with a partner. Write calculations with units and explain your decisions. Complete the exit check individually. Calculators allowed.

1. Observe two windows

A weather-station reader keeps one copy of each station record it needs. Every record has a 64 B payload. Count payload only; exclude overhead.

Time	Requests in order
10:00 to before 10:01	S2, S5, S2, S8
10:01 to before 10:02	S5, S3, S5, S3
10:02 to before 10:03	S3, S9, S3, S9

1a. For 10:00 to before 10:02, write the working set W, its cardinality |W|, and the payload budget B. Show the multiplication.

1b. Shift the window to 10:01 to before 10:03. Write its W, |W|, and B. Show the multiplication.

1c. Both windows contain eight requests. Which records leave W, which enter W, and why does B change?

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2. Test each tier

The classroom machine has 16 MB cache and 16 GB RAM. Each station record has a 64 B payload. Each job below supplies its own distinct-record count, separate from the short log.

B = |W| × record_size. Fits(t) is true exactly when B ≤ capacity(t). Count payload only; exclude overhead.

1 MB = 1,000,000 B; 1 GB = 1,000,000,000 B; 1 GiB = 2^{30} B. All exercise capacities use decimal units.

2a. Job A needs 250,000 distinct records. Compute B in bytes and MB. Write both capacity comparisons and decide Fits(cache) and Fits(RAM).

2b. Job B needs 250,000,001 distinct records. Compute B in bytes. Write both capacity comparisons and decide Fits(cache) and Fits(RAM).

2c. A partner says, “If a job fits cache, all its records are already there and every request will be fast.” Does the capacity test support this claim? Explain what it establishes.

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3. Budget a complete model run

Qwen3.5-9B weights occupy 19,306,216,416 B. This exercise gives the GPU its own 24,000,000,000 B budget. The machine also has 16 GB RAM.

Allocation in GPU memory	**Bytes**
Weights	19,306,216,416 B
Other runtime allocations	1,200,000,000 B
Proposed context/KV cache	3,600,000,000 B

For this exercise, these are all the allocations in the run. The runtime and context sizes are supplied assumptions.

3a. Add all three allocations. Decide whether this proposed run fits the GPU. Show the comparison in bytes.

3b. Keep weights and other runtime allocations fixed. What is the maximum context/KV cache budget that fits? Show the subtraction in bytes.

3c. A second proposal uses 3,200,000,000 B for context/KV cache. Compute the new total. Does it fit? If so, how many GPU bytes are unused? Explain whether unused RAM can increase this exercise's GPU capacity.

Will it fit?

Will this data fit in memory?

4. Explain the hierarchy

A job's payload is 192 MB. The classroom machine has 16 MB cache and 16 GB RAM.

4. A partner proposes making cache as large as RAM and dropping RAM. Explain the capacity/latency tradeoff this proposal overlooks. Which tier holds the job's records when cache cannot hold them all, and how does that affect the wait for those records?

5. Exit check

Name: _____

Answer individually.

5a. A window gains 100 more requests, all for station records already in W. Does |W| change? Explain why.

5b. A model's weights leave 4 GB free on the GPU. Context/KV cache needs 2.5 GB and other runtime allocations need 2 GB. A partner says each fits the remaining 4 GB, so the whole run fits. Is the partner right? Explain.